

LAB 10

Inter-VLAN Routing Using Router-on-a-Stick

Scenario

A company has two departments: **Development (DEV)** and **Human Resources (HR)**. To improve network management and security, the administrator wants to separate them into two VLANs. DEV uses **VLAN 10**, while HR uses **VLAN 20**. Each department has two PCs connected to a switch. A router is used to allow communication between the two VLANs through **Inter-VLAN Routing**.

Requirements

- 1 Router
- 1 Switch
- 4 PCs
- VLAN 10 → DEV
- VLAN 20 → HR

IP Addressing

Department	VLAN	Network
DEV	10	192.168.10.0/24
HR	20	192.168.20.0/24

Tasks

1. Create the topology in Cisco Packet Tracer.
2. Create VLAN 10 and VLAN 20.
3. Assign PCs to their respective VLANs.
4. Configure the switch-router link as a trunk.
5. Configure router subinterfaces for both VLANs.
6. Assign IP addresses and default gateways.
7. Test connectivity between VLANs using ping.

Expected Result

PCs from VLAN 10 and VLAN 20 should communicate successfully through the router.